📄 **SMART-DPR**

(Under e-Nirikshan – Drishti AI Protocol)

Project Reference

KERALA-IFRA-2026-PWD-772

Work Name

Development of Multi-Level Smart Parking & Commercial Hub
Ernakulam North

Tender Value

₹42,50,00,000 (Forty-Two Crores Fifty Lakhs Only)

Contractor

Apex Build-Tech Solutions Ltd.

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1. TECHNICAL EXECUTIVE SUMMARY

This Smart-DPR presents the execution strategy for the Ernakulam Smart Parking & Commercial Hub project.

Apex Build-Tech proposes a 240-day accelerated timeline using Pre-cast Concrete Technology to:

- Minimize traffic disruption
- Optimize vertical construction
- Reduce curing cycle delays
- Enable milestone-based AI validation

This proposal is structured for automated feasibility analysis under e-Nirikshan.

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2. ARCHITECTURAL & ENGINEERING SPECIFICATIONS

Total Built-up Area: 45,000 Sq. Ft.
Configuration: G+4 Structure
Structural Type: RCC Framed with Post-Tensioned Slabs

Concrete Grade: M40
Steel Grade: FE550

Sustainability Features

- 50kW Rooftop Solar Plant
- 20,000 Liter Rainwater Harvesting
- Smart IoT Lighting & Energy Monitoring

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3. PHASED EXECUTION TIMELINE (WBS)

Phase	Description	Timeline
P1	Site Preparation	Day 1–15
P2	Foundation & Piling	Day 16–75
P3	Vertical Construction (G+4)	Day 76–180
P4	MEP & Smart Integration	Day 181–220
P5	Finishing & Handover	Day 221–240

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4. RESOURCE DEPLOYMENT PLAN

Human Resources

- On-Site Engineers: 04
- Safety Officers: 02
- Skilled Labor: 45
- Unskilled Labor: 80
- Peak Workforce: 131 Personnel

Machinery Deployment

- Tower Crane (12T): 1 Unit
- Transit Mixers: 4 Units
- IoT-Enabled Concrete Testers: 2 Units
- Excavators: 3 Units

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5. FINANCIAL ESTIMATION

Expense Head	Amount (Cr)	%
Civil Materials	22.40	52.7%
Labor & Overhead	8.10	19.1%
Technology & MEP	6.20	14.6%
Contingency	2.12	5.0%
Contractor Profit	3.68	8.6%

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6. TRANSPARENCY COMPLIANCE (e-Nirikshan Protocol)

The Contractor agrees to:

1. Weekly 360° Visual Uploads
2. IoT Telemetry Integration
3. Geo-Fenced Labor Attendance
4. AI-Verified Concrete Strength Logs
5. Digital Milestone Lock-In